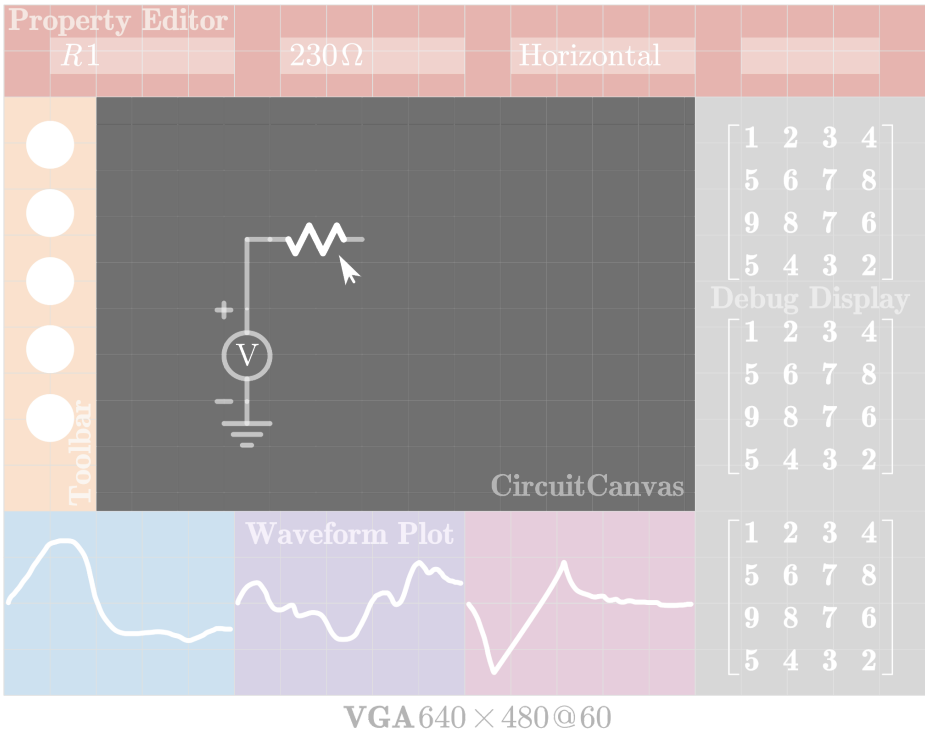


I-FDP PROPOSAL DOCUMENT

Group ID: S1_G9	Description (Pictures / Drawings / Sketches / Gantt Charts are optional and not necessary)
Group Theme:	 VGA 640 × 480 @ 60 <i>Fig 1 Proposed GUI Layout</i> Our group proposes to implement an interactive FPGA-based analog circuit simulator on the Basys3 board using Verilog. Rather than building a conventional keypad-style calculator, we aim to create a “calculator” in which the user constructs simple analog circuits and observes the computed electrical behaviour in real time. The system will support interactive circuit drawing on a VGA canvas using a mouse. Users will be able to place and edit components such as resistors, independent voltage sources, independent current sources, and linear controlled sources. A property editor will allow users to modify component labels and values, with support for simple arithmetic expressions during value entry. At the backend, the design will (1) generate a circuit netlist, (2) construct the corresponding system matrix using stamping methods, and (3) solve it using single-precision floating-point computation. The simulator is planned to support DC analysis first, with further enhancement toward AC/transient-style waveform display if time permits. The design will include a waveform plotting area, debug/status displays, and optional visual overlays such as voltage colour maps and current particle animation. The 96×64 OLED will be used as a compact state indicator. Basys3 switches and LEDs will be used for quick toggles such as enabling overlays, switching display modes, or resetting the interface.
Group member 1: Chen Shimin	Member Responsibility: Shimin will take responsibility for the core system architecture and backend computation pipeline (matrix calculation). The work includes the circuit canvas interaction, netlist generation, matrix generation through MNA-style stamping, and the matrix solver core. He will also coordinate the interfaces between frontend editing actions and backend recomputation so that the simulator behaves synchronously. In addition, he will support final system integration and help define common module interfaces. Week 8 Task: Define a storage format for electrical components and display on the front end./ Establish interfaces for the renderer, editor, netlist generator, and solver core Week 9 Task: Implement netlist generation using a flooding/connectivity algorithm/ Integrate front-end editing result and back-end calculation Week 10 Task: Continue development of single-precision floating-point matrix solver core/ Verify correct solver behaviour with reference software results Week 11 Task: Integrate solver pipeline with frontend display path/ Debugging and Integrating Week 12 Task: Final integration, optimization, and bug fixing/ Whole system testing with software

Group member 2: Hsu Yuchen	Member Responsibility: YUCHEN will mainly handle the waveform plot, matrix/state display, and analysis control support, including the linearization/control flow around analysis execution. His work helps expose the “calculator” result into a graphical form: not only numeric node values, but also waveform-style plots and internal matrix displays for debugging. He will also help manage how analysis results are packaged and forwarded to the display subsystem. Week 8 Task: Define the VGA communication and display different colors on the external screen Week 9 Task: Implement initial waveform plotting pipeline using stored sampled values/ Begin matrix display module for showing solver-related numerical information Week 10 Task: Integrate waveform plot with backend-generated values Week 11 Task: Refine plotting behaviour and mode switching between graph/debug/canvas views/ Test end-to-end result visibility with real example circuits Week 12 Task: Support the final display and validate the plotting result
Group member 3: Wang Zhirui	Member responsibility: Zhirui will focus on the frontend editing experience, especially the property editor and the visual overlay system. This includes creating widgets for displaying and editing component properties and implementing visual effects such as highlighting circuits and current particle animation on the VGA canvas. His role is important for transforming the project from a backend solver into a user-friendly and visually intuitive graphical calculator. He will also help connect simulation results to visualization modules so that solved values can be updated instantaneously when updates are made by the user. Week 8 Tasks: Implemented PS2-equivalent protocol to enable mouse interaction with FPGA board through the USB-A I/O port / Design the features of the property editor that include editable features, data path and visualization styles. Week 9 Tasks: Implement a property editor interface and an interactive UI that synchronizes / changes with the backend system. / Study voltage/current visualization styles suitable for FPGA rendering Week 10 Tasks: Extend the property editor to support value updates from virtual keyboard input / Implement and test the current particle or direction animation overlay prototype Week 11 Tasks: Integrate editor output and current visualization with backend-updated circuit data / Refine readability and responsiveness of property editing Week 12 Tasks: Polish the UI, fix rendering glitches, and improve clarity of visual feedback / Perform integration testing with full circuit workflow and demo preparation
Group member 4: Xie Ziyi	Member Responsibility: Responsible for developing the text/value input and debugging interface. Specifically, this includes the keyboard input subsystem (displayed at the bottom-left of the PC interface) and the debug information display on the OLED screen. Finally, I will ensure that my input and debugging systems integrate smoothly with other modules. Week 8 Tasks: I have finished the basic keyboard input handling module on the OLED screen. I have implemented a key-mapping mechanism that translates button presses into corresponding numeric or unit inputs. Week 9 Tasks: I will implement a stable keyboard input pipeline for entering component values on the PC screen. I will start building the initial debug display module on the OLED screen to indicate circuit issues such as open circuit, short circuit, and missing ground. Week 10 Tasks: I will extend the keyboard module to support simple arithmetic expressions for component parameters (e.g., addition and subtraction of integers). I will improve input validation and error handling for invalid entries. I will expand the debug interface to display system status. Week 11 Tasks: I will integrate the keyboard input system with the property editor and component selection workflow. I will refine the debug display design for system integration and presentation readiness. Week 12 Tasks: I will conduct reliability testing for input operations and system debugging. I will perform iterative refinements to improve stability and usability. I will finalize the user interaction flow, support system cleanup for the final demonstration, and contribute to project documentation.
Group Integration:	Week 10 integration: Integrate the circuit canvas interaction, keyboard/property input path, netlist generation, and basic matrix stamping/solver pipeline for simple DC circuit testing. Week 11 integration: Integrate solver outputs with VGA visualization modules, including property updates, waveform/debug display, and voltage/current overlay control. Week 12 integration: Integrate the full system into a stable final demo, including UI control, circuit editing, backend recomputation, display modes, OLED/LED/switch status indication, and reliability testing